

```
In [1]: print("3rd April 2026 Class on ...")
```

3rd April 2026 Class on ...

```
In [3]: # Check even or odd number
x = 5
r = x%2
if r==0:
    print("Even Number ")
else:
    print("odd Number ")
```

odd Number

```
In [4]: # Check even or odd number
x = 5
r = x%2
if r==0:
    print("Even Number ")
else:
    print("odd Number ")
if r==2:
    print("Even Number ")
if r==3:
    print("odd Number ")
```

odd Number

```
In [7]: #Nestedif

x = 14
r = x%2
if r==0:
    print("Even Number ")
    if x>5:
        print("Grater Number ")
    else:
        print("Lesser Number ")

else:
    print("odd Number ")
```

Even Number

Grater Number

```
In [10]: # If elif else

x = 1

if x==1:
    print("ONE ")
if x==2:
    print("TWO ")
if x==3:
    print("Three ")
```

ONE

```
In [13]: # If elif else
```

```
x = 3

if x==1:
    print("ONE ")
elif x==2:
    print("TWO ")
elif x==3:
    print("Three ")
elif x==4:
    print("Four ")
```

Three

```
In [15]: # If elif else

x = 10

if x==1:
    print("ONE ")
elif x==2:
    print("TWO ")
elif x==3:
    print("Three ")
elif x==4:
    print("Four ")
else :
    print("Number not found")
```

Number not found

```
In [16]: age = 20
if age >= 18:
    print("Eligible to vote.")
```

Eligible to vote.

```
In [17]: age = 19
if age > 18: print("Eligible to Vote.")
```

Eligible to Vote.

```
In [18]: age = 10
if age <= 12:
    print("Travel for free.")
else:
    print("Pay for ticket.")
```

Travel for free.

```
In [19]: marks = 45
res = "Pass" if marks >= 40 else "Fail"
print(f"Result: {res}")
```

Result: Pass

```
In [ ]: # practice from below url
# https://www.geeksforgeeks.org/python/conditional-statements-in-python/
```

```
In [41]: age = 70
is_member = True

if age >= 60:
```

```

    if is_member:
        print("30% senior discount!")
    else:
        print("20% senior discount.")
else:
    print("Not eligible for a senior discount.")

```

30% senior discount!

```

In [24]: age = 25

if age <= 12:
    print("Child.")
elif age <= 19:
    print("Teenager.")
elif age <= 35:
    print("Young adult.")
else:
    print("Adult.")

```

Young adult.

```

In [25]: ***LOOPS**

```

```

In [32]: i = 1 # initialise

while i<=5: # Condition
    print("data science")
    i = i+1 # increment

```

data science
data science
data science
data science
data science

```

In [33]: i = 1 # initialise i =1

while i<=5: # Condition 1<5, 2<5, 3<5, 4<5, 5<=5, 6<=5 exit
    print("data science")
    i = i+1 # increment 1+1 =2 ,2,3,4,5,6.

```

data science
data science
data science
data science
data science

```

In [34]: i = 1 # initialise i =1

while i<=5: # Condition 1<5, 2<5, 3<5, 4<5, 5<=5, 6<=5 exit
    print("data science",i)
    i = i+1 # increment 1+1 =2 ,2,3,4,5,6.

```

data science 1
data science 2
data science 3
data science 4
data science 5

```
In [35]: i =5 # initialise i =1

while i>=1: # Condition 1<5, 2<5, 3<5, 4<5, 5<=5, 6<=5 exit
    print("data science",i)
    i = i-1
```

```
data science 5
data science 4
data science 3
data science 2
data science 1
```

```
In [ ]: ### Nested while - while inside while called nested while
```

```
In [36]: i =1 # initialise i =1

while i<=5: # Condition 1<5, 2<5, 3<5, 4<5, 5<=5, 6<=5 exit
    print("data science",i)
    j=1
    while j<=4:
        print('technology')
        j=j+1
    i = i+1 # increment 1+1 =2 ,2,3,4,5,6.
    print()
```

```
data science 1
technology
technology
technology
technology

data science 2
technology
technology
technology
technology

data science 3
technology
technology
technology
technology

data science 4
technology
technology
technology
technology

data science 5
technology
technology
technology
technology
```

```
In [38]: i =1 # initialise i =1

while i<=5: # Condition 1<5, 2<5, 3<5, 4<5, 5<=5, 6<=5 exit
    print("data science",i, end="")
```

```

j=1
while j<=4:
    print('technology',end="")
    j=j+1
i = i+1 # increment 1+1 =2 ,2,3,4,5,6.
print()

```

```

data science 1technologytechnologytechnologytechnology
data science 2technologytechnologytechnologytechnology
data science 3technologytechnologytechnologytechnology
data science 4technologytechnologytechnologytechnology
data science 5technologytechnologytechnologytechnology

```

```

In [39]: i =1
while i<=2:
    j=0
    while j<=2:
        print(i*j,end="")
        j+=1
    print()
    i+=1

```

```
012
```

```
024
```

```

In [40]: i =1
while i<=4:
    j=0
    while j<=3:
        print(i*j,end="")
        j+=1
    print()
    i+=1

```

```
0123
```

```
0246
```

```
0369
```

```
04812
```

```
In [ ]:
```